

SlimTorq is a closed-loop simulator for a slotless PMSM driven by Field-Oriented Control. The plant is an OpenModelica FMU (`SlotlessPMSM_abc.fmu`); the FOC controller, encoder, PWM inverter, and optional LCL filter run in Python and update the FMU at the inner sim step. Configure a run from the left panel, click **Simulate**, read the plots on the right. Every Simulate click runs a fresh sim and overwrites `.temp/<family>_<variant>.parquet`.

INSTALL & RUN (DOCKER, ONE CALL)

Only requirement on the host is Docker Engine or Docker Desktop (docs.docker.com/get-docker). The published image bundles the Python deps, motor catalog, assets, and the pre-built FMU — no Python or OpenModelica toolchain needed.

1. Install Docker on your PC (Linux, macOS, or Windows + WSL2).
2. From the repo root run: `docker compose up`
3. Open `http://localhost:8000` in your browser.

Compose pulls (or builds) the image, mounts a named volume for the parquet artifacts, and starts the service. All runtime settings — port, CORS, frontend bundle path, artifact directory, catalog path — live in `config/app.yaml` and can be overridden per-env with the double-underscore convention (e.g. `APP_PORT=9000`, `ARTIFACTS_DIR=/var/data`).

CONFIG PANEL — WHAT EACH FIELD DOES

Motor

variant SlimTorq catalog variant. R_s , L_s , ψ_m , p , J , V_{dc} are derived from this — no manual override.

Power stage

f_{pwm} PWM carrier frequency (Hz). Sets the inner sim step ($T_{pwm}/20$ by default) and the current-loop bandwidth $\approx f_{pwm}/9.4$.

t_{dead} Gate-driver dead time (s). 0 disables. Adds 5th/7th harmonic distortion to i_{abc} .

PWM mode `sine`, `svpwm`, `dpwmmax`, `dpwmmin`, `dpwm1`, or `auto` (svpwm $\leftrightarrow$ dpwm1 at $m_i = 0.45/0.55$). DPWMx clamps one phase to a rail, ~33% less switching loss.

Output filter (LCL)

enable Insert LCL low-pass between inverter terminals and motor.

f_c Target cutoff. L_f , C_f , R_d derived as `$L_f=L_s/4$` , `$C_f=1/(\omega_c^2 \cdot L_f)$` , `$R_d=\sqrt{(L_f/C_f)/3}$` . Hint line shows recommended bounds.

Encoder (inductive flux model)

N_{bits} Absolute resolution; $LSB = 2\pi/2^N$.

θ_{offset} Constant mounting offset (rad).

A_k , k_k , ϕ_k Three cyclic harmonics: $\theta_{err} = \sum A_k \cdot \sin(k_k \cdot \theta + \phi_k)$.

$T_{s,enc}$ Encoder sample period (s). ZOH between samples. Visible as sawtooth $\omega \cdot T_{s,enc}$ on the encoder-error plot.

Trajectory

t_{end} , t_{step} Sim horizon and load-step time (s).

T_L^{ref}/T_e^{peak} Step amplitude as fraction of catalog peak torque.

T_f Run-length override; *None* = auto.

Timing

Δt_{sim} Inner sim step (s); *None* = $T_{pwm}/20$. Must $\leq T_{pwm}/10$ (enforced). Use $\leq T_{pwm}/50$ for clean switching-mode results.

Current loop — PI tuning

Each rule is *plant-aware*: with the LCL filter off the plant is treated as the LR motor (R_s , L_s); with the LCL filter on it switches to the low-frequency equivalent $L_{eq}=L_1+L_{load}$, $R_{eq}=R_1+R_{load}$, and rejects the tuning if the implied closed-loop bandwidth gets too close to the LCL resonance ($\omega_{bw} \leq \omega_{res}/10$, conservative) or to $f_{pwm}/10$.

Modulus Optimum $K_p=L \cdot f_{pwm}/3$, $K_i=R \cdot f_{pwm}/3$. Default. $\approx 63^\circ$ PM, 4.3 % overshoot.

Skogestad SIMC $K_p=L/(T_c+\tau)$, $T_i=\min(L/R, k_1 \cdot (T_c+\tau))$. $T_c=$ *None* uses $\tau=1.5/f_{pwm}$; $k_1=1.44$ default $\rightarrow \sim 3\times$ sharper load-step rejection.

Manual Type K_p/K_i directly. Bypasses the LCL safety gate.

Debug section

Inverter mode `ideal` = v_{abc}^{ref} straight to FMU; `average` = duty calc but cycle-averaged voltage (no dead-time, isolates scaling bugs); `switching` = real PWM compare + dead-time (default).

Sticky controls. Config panel stays pinned while the plot column scrolls. The **Simulate** button sticks to the bottom of the config panel so it's always reachable. Hard-refresh after CSS edits if the page looks stale.

12 PANELS · WHAT TO LOOK FOR

The plot column renders 12 Plotly figures from the most-recent parquet. Each figure pairs a measurement (solid) with its reference (dashed) using the same palette slot per signal. Steady-state metrics are computed on the trailing 80 % of samples to skip the load-step transient.

Tracking

Torque (T_L^{ref} vs T_e), i_q^* vs i_q , i_d^* vs i_d . Each subplot title carries the steady-state RMS reference-tracking error in %.

PI performance

e_d , e_q errors; v_d^{ref} , v_q^{ref} ; $|v_{dq}|$ vs the $V_{\text{max}} = V_{\text{dc}}/2$ vector limit (red-dotted). PI-saturation rug strip below shows when the integrator was frozen.

dq round-trip

$v_d^{\text{actual}} - v_d^{\text{ref}}$, $v_q^{\text{actual}} - v_q^{\text{ref}}$, reconstructed from logged post-inverter v_{abc} via Clarke+Park. **Cycle-average ≈ 0** proves inverter scaling is correct. Switching-mode shows ripple; average/ideal modes show flat zero.

 i_q zoom

$5 \times T_{\text{pwm}}$ window in steady state with carrier-valley guides. Shows the controller's actual measurement noise vs reference.

 i_q FFT, ω FFT, i_a FFT

Log-magnitude spectra (trailing 80 %, DC removed). Switching ripple shows at $2 \cdot f_{\text{pwm}}$ in i_q , at $f_{\text{pwm}} \pm n \cdot f_1$ in i_a . Dead-time signature: $5^{\text{th}}/7^{\text{th}}$ of the electrical fundamental in i_a .

Phase currents (abc)

i_a , i_b , i_c overlaid. With switching enabled and no filter, ripple amplitude $\Delta i_{\text{pp}} \approx V_{\text{dc}} \cdot T_{\text{pwm}} / (8 \cdot L_s)$.

Phase voltages

v_a^{ref} dashed, post-inverter v_a faint solid, and (when LCL on) v_a^{motor} at filter output as medium solid. With filter on, the motor trace should hug the smooth ref while the inverter trace hits $\pm V_{\text{dc}}/2$.

PWM duties

d_a , d_b , d_c . With `svpwm` / `dpwm` the shape clearly departs from pure sine — visual confirmation the zero-sequence injection took effect.

Encoder error

$\text{wrap}(\theta_m^{\text{true}} - \theta_m^{\text{meas}})$ in mrad. Dashed envelope at $\omega \cdot T_{\text{s,enc}}$ is the ZOH bound; the trace should ride inside this band. Excursions outside it are cyclic harmonics + quantization + offset.

Speed + saturation

ω_m^{true} vs ω_m^{meas} , plus a rug strip marking d/q PI saturation events.

Observer (LCL only)

Appears next to *Signal Processing* when the LCL filter is enabled. Two panels overlay the measured dq currents (i_d^{meas} , i_q^{meas}) with the Luenberger observer's three estimated LCL states (i_1 , v_c , i_m) and the derived capacitor current $i_c = i_1 - i_m$. The accompanying FFT shows where i_c 's spectral energy sits — peaks near the LCL resonance are what future *active damping* $v_{\text{damp}} = -K_d \cdot i_c$ will target.

WORKFLOW TIPS

- **Diagnose noise:** bypass with `Inverter mode = ideal`. If clean, the inverter is the source. Then try `average` — non-zero `mean  $\Delta v_d$` in the dq round-trip means a scaling bug; flat zero means switching ripple is the issue.
- **Tune for disturbance rejection:** switch PI to Skogestad, lower k_1 toward 1.44. Tighter integrator → smaller `err %` in the tracking subplot titles.
- **Pick f_c for the LCL:** aim near $f_{\text{pwm}}/10$ for ~40 dB carrier attenuation. The hint under the input gives the loop-safe floor for the current f_{pwm} .
- **Sample-rate aliasing check:** at fixed f_{pwm} , halve Δt_{sim} . If noise drops, it's edge quantization in the modulator, not real inverter physics.
- **Output:** each run writes `.temp/<family> <variant>.parquet` with full per-sample log and `slimtorq.*` metadata (params hash, tuning, modes, filter values).